

| Algorithm / Data Structure | Key Operation              | Best Time     | Average Time                   | Worst Time                     | Space Complexity |  |
|----------------------------|----------------------------|---------------|--------------------------------|--------------------------------|------------------|--|
| Array                      | Access                     | $O(1)$        | $O(1)$                         | $O(1)$                         | $O(n)$           |  |
| Array                      | Search                     | $O(1)$        | $O(n)$                         | $O(n)$                         | N/A              |  |
| Array                      | Insert (general)           | $O(n)$        | $O(n)$                         | $O(n)$                         | N/A              |  |
| Array                      | Delete (general)           | $O(n)$        | $O(n)$                         | $O(n)$                         | N/A              |  |
| Linked List (Singly)       | Access                     | $O(n)$        | $O(n)$                         | $O(n)$                         | $O(n)$           |  |
| Linked List (Singly)       | Search                     | $O(1)$        | $O(n)$                         | $O(n)$                         | N/A              |  |
| Linked List (Singly)       | Insert (general w/ search) | $O(1)$        | $O(n)$                         | $O(n)$                         | N/A              |  |
| Linked List (Singly)       | Delete (general w/ search) | $O(1)$        | $O(n)$                         | $O(n)$                         | N/A              |  |
| Stack (LIFO)               | Push (efficient)           | $O(1)$        | $O(1)$                         | $O(1)$                         | $O(n)$           |  |
| Stack (LIFO)               | Pop (efficient)            | $O(1)$        | $O(1)$                         | $O(1)$                         | N/A              |  |
| Queue (FIFO)               | Enqueue (efficient)        | $O(1)$        | $O(1)$                         | $O(1)$                         | $O(n)$           |  |
| Queue (FIFO)               | Dequeue (efficient)        | $O(1)$        | $O(1)$                         | $O(1)$                         | N/A              |  |
| Heap (Max-Heap)            | Insert                     | $O(\log n)$   | $O(\log n)$                    | $O(\log n)$                    | $O(n)$           |  |
| Heap (Max-Heap)            | Delete Max                 | $O(\log n)$   | $O(\log n)$                    | $O(\log n)$                    | N/A              |  |
| BST                        | Search                     | $O(1)$        | $O(\log n)$                    | $O(n)$                         | $O(n)$           |  |
| BST                        | Insert                     | $O(1)$        | $O(\log n)$                    | $O(n)$                         | N/A              |  |
| BST                        | Delete                     | $O(\log n)$   | $O(\log n)$                    | $O(n)$                         | N/A              |  |
| AVL Trees                  | Search                     | $O(\log n)$   | $O(\log n)$                    | $O(\log n)$                    | $O(n)$           |  |
| AVL Trees                  | Insert                     | $O(\log n)$   | $O(\log n)$                    | $O(\log n)$                    | N/A              |  |
| AVL Trees                  | Delete                     | $O(\log n)$   | $O(\log n)$                    | $O(\log n)$                    | N/A              |  |
| Bubble Sort                | Sort comparisons           | $O(n)$        | $O(n^2)$                       | $O(n^2)$                       | $O(1)$           |  |
| Selection Sort             | Sort comparisons           | $O(n^2)$      | $O(n^2)$                       | $O(n^2)$                       | $O(1)$           |  |
| Insertion Sort             | Sort comparisons           | $O(n)$        | $O(n^2)$                       | $O(n^2)$                       | $O(1)$           |  |
| Merge Sort                 | Sort comparisons           | $O(n \log n)$ | $O(n \log n)$                  | $O(n \log n)$                  | $O(n)$           |  |
| Hashing (chaining)         | Search (collisions)        | $O(1)$        | $O(1)$                         | $O(n)$                         | $O(n)$           |  |
| BFS (Graph Traversal)      | Traversal                  | $O(V+E)$      | $O(V+E)$                       | $O(V+E)$                       | $O(V)$           |  |
| DFS (Graph Traversal)      | Traversal                  | $O(V+E)$      | $O(V+E)$                       | $O(V+E)$                       | $O(V)$           |  |
| Spanning Tree              | Create (general ranges)    | Varies        | $O(E \log V)$ to $O(E \log E)$ | $O(E \log V)$ to $O(E \log E)$ | Varies           |  |